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Roll No:.....

**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI***(An ISO 9001:2008 Certified College)***B. TECH DEGREE FIRST INTERNAL – October-2023****SEVENTH SEMESTER****MCN401 – INDUSTRIAL SAFETY ENGINEERING***(Common to all branches)***Time: 2 hr****Max. Marks: 50****PART A**

Answer all questions (3x5=15 Marks)

Qn. No.	Question	Mark	K L	CO&KL
✓1	What is the need for safety in a work environment?	3	K2	CO1 (K2)
✓2	Differentiate between unsafe act and unsafe condition.	3	K2	CO1 (K2)
✓3	What are the five S (5S) of Housekeeping?	3	K1	CO2 (K2)
✓4	Define safety activity rate, severity rate and frequency rate.	3	K1	CO2 (K2)
✓5	What are the hazards related to scaffolds?	3	K1	CO3 (K2)

PART B (7 x 5 = 35 Marks)**Module 1: Qn. No. 6,7 and 8 – Answer any two**

✓6	Explain the responsibilities of a safety officer in the implementation of safety in an industry.	7	K2	CO1 (K2)
7	What are the various accident causation theories. List all of them and explain any two.	7	K2	CO1 (K2)
8	Explain the responsibilities of the government towards safety in an organization.	7	K2	CO1 (K2)

Module 2: Qn. No. 9, 10, and 11 – Answer any two

✓9	List the various <u>respiratory PPEs</u> . Explain any two in detail.	7	K2	CO2 (K2)
10	What is a confined space? What are the safety precautions to be taken before entering it. Explain the role of the standby person?	7	K2	CO2 (K2)
✓11	What are the major objectives of work permit in a hazardous work site?	7	K2	CO2 (K2)

Module 3: Mandatory Question

✓12	Explain the various safety provisions mentioned in the NBC (National Building code) for construction?	7	K2	CO3 (K2)
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B. TECH DEGREE SECOND INTERNAL TEST - December-2023
SEVENTH SEMESTER
MCN401 - INDUSTRIAL SAFETY ENGINEERING
(Common to all branches)

Time: 2 hr

Max. Marks: 50

PART A

Answer all questions (3x5=15 Marks)

Q.No.	Question	Mark	KL	CO & KL
1	List the various hazards of underwater works.	3	K2	CO3 (K2)
2/	How can organizations effectively mitigate the risks associated with manual material handling to ensure both employee well-being and operational efficiency?	3	K2	CO4 (K2)
3	What are the key elements that should be included in a comprehensive hearing conservation program to safeguard the auditory health of employees in noisy work environments?	3	K2	CO4 (K2)
4/	Why is the provision and utilization of Material Safety Data Sheets (MSDS) crucial within the chemical industry?	3	K2	CO5 (K2)
5/	Explain the hierarchy of control of chemical hazards.	3	K2	CO5 (K2)

PART B

Answer five questions (7x5=35 Marks)

Module 4: Qn. No. 6,7 and 8 - Answer any two
 Module 5: Qn. No. 9,10 and 11 - Answer any two
 Module 3: Qn. No. 12 is mandatory

Q. No	Question	Mark	KL	CO & KL
6	Enumerate the diverse categories of machine guarding devices employed across industries, and discuss on the appropriateness of each device in ensuring safety.	7	K2	CO4 (K2)
7	Discuss the safety issues associated with gas welding operations.	7	K2	CO4 (K2)
8	Explain the potential hazards associated with grinding operations	7	K2	CO4 (K2)
9	Explain the need for a Preliminary Hazard Analysis in a hazardous industry	7	K2	CO5 (K2)
10	What role does Hazard and Operability Analysis (HAZOP) play in ensuring safety, and what are the steps involved in conducting a HAZOP analysis?	7	K2	CO5 (K2)
11	How are fires classified, and what are the different types of fire extinguishers utilized in industrial settings to combat specific fire hazards?	7	K2	CO5 (K2)
12	Identify and discuss the important aspects of construction safety provisions in National Building Code.	7	K2	CO3(K2)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 scheme)

Course Code: MCN401**Course Name: INDUSTRIAL SAFETY ENGINEERING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- ✓ 1 With suitable examples differentiate between unsafe act and unsafe conditions referred to construction industry. (3)
- ✓ 2 How do voluntary agencies supports safety measures in various organizations? (3)
- ✓ 3 With suitable sketches mention the features of any two PPE for eye protection in industrial hazardous sites. (3)
- ✓ 4 Explain the major objectives of work permit systems. (3)
- ✓ 5 Explain any five critical reasons for high rate of accidents in construction industry. (3)
- ✓ 6 As an engineering supervisor how do you ensure the overall safety during demolition operations? (3)
- ✓ 7 Explain the key features of hearing conservation program in production industries. (3)
- ✓ 8 Explain the safety guidelines to be followed for the lifting hooks used in heavy industrial mechanisms. (3)
- ✓ 9 Explain the different causes of electric hazards. (3)
- ✓ 10 Write short note on toxic gas release due to chemical and fire explosion. (3)

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- ✓ 11 a) Explain Heinrich's domino theory for accident causation. Also mention any two modifications made on this basic concept. (7)
- b) Explain the six distinguished functions of safety committee in an industry. (7)

OR

- ✓ 12 a) With suitable examples explain how does safety management improve productivity of a firm. (6)
- b) Explain the responsibilities of supervisors to ensure safety in an industrial organization. (8)

Module II

- 13 a) Explain the key functions of any three different types of respiratory protective equipments. (6)
- b) Explain any ten important responsibilities of housekeeping head of an organization. (8)

OR

14

Write short notes on the following:

(14)

- i. Cold work permit
- ii. Reactive measures for safety performance monitoring
- iii. Five S (5S) concept in housekeeping.

Module III

15

Write short notes on the following:

(14)

- i. Cumulative Trauma Disorders due to ergonomic issues in work sites
- ii. Hazards and risks during Underpinning works
- iii. Safety measures during excavation works

OR

16

- a) With suitable schematics, mention any eight specific safety precautions for scaffolding to ensure the safety of the workers. (9)
- b) Explain the classification of residential buildings based on the National Building Code of India (5)

Module IV

17

- a) Describe about the safety practices to be followed during arc welding process. (8)
- b) Differentiate between manual and mechanised material handling with suitable examples. (6)

OR

18

- a) An industrial plant is equipped with crane, conveyors, hoist etc. Explain the maintenance strategy to be followed for any of the material handling equipment to ensure safe working environment. (9)
- b) With suitable schematics, explain the operation of any presence sensing device suitable as a safety guarding device for an industrial unit. (5)

Module V

19

- a) Describe the relevance and procedure for HAZOP for safety assessment. (9)
- b) Explain how does MSDS differ from a product label referred to material safety. (5)

OR

20

- a) Describe about the classification of fires. Also explain any four types of fire extinguishers. (9)
- b) Describe about any four hazardous properties of chemical materials. (5)
